

SIEM Implementation (Wazuh)

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1. Introduction & Objectives

Project Context

In the modern threat landscape, effective security requires continuous monitoring to identify indicators of compromise (IOCs) and indicators of attack (IOAs) across heterogeneous systems. A **SIEM** system fulfills this requirement by centralizing log data, normalizing events, and applying advanced correlation rules to identify security incidents that would otherwise be lost in raw log volume. This project focuses on building such a system.

Project Goal

The primary goal was to implement a fully functional, small-scale SIEM lab suitable for real-world threat detection, focusing on the principles of reliability and comprehensive coverage.

Specific Objectives

- **Deployment:** Successfully deploy the three core components of the Wazuh platform: the Manager, Indexer, and Dashboard.
- **Endpoint Integration:** Establish stable, bidirectional communication with a production-like endpoint (Windows 10).
- **Correlation Demonstration:** Prove the ability to elevate low-severity events into high-severity security incidents (e.g., Brute Force attack).
- **Vulnerability Detection:** Verify the system's capability to assess and alert on critical software vulnerabilities.
- **Documentation & Troubleshooting:** Provide a complete log of all installation and advanced troubleshooting steps undertaken to achieve operational stability, showcasing deep debugging expertise.

2. System Architecture

The implemented SIEM solution utilizes a standard client-server architecture with Wazuh components.

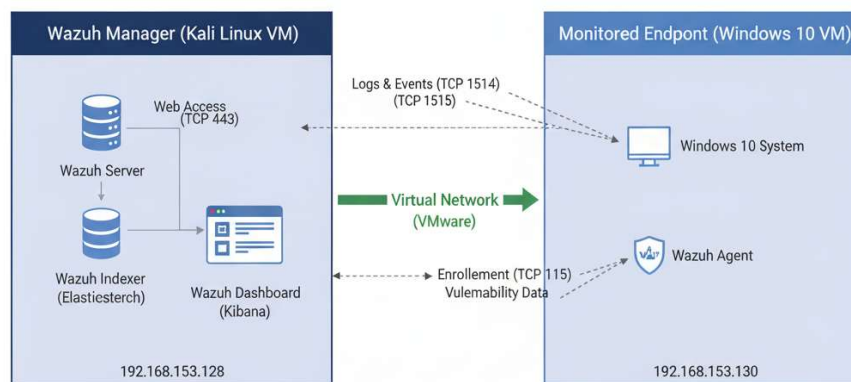
Tools Used

- **SIEM Platform: Wazuh (v4.14.0):** A comprehensive, open-source platform that combines HIDS, SIEM, and XDR capabilities into a single solution.
- **Manager & Server OS: Kali Linux VM:** Used as the deployment host for the Wazuh Manager, Indexer (Elasticsearch), and Dashboard (Kibana/Wazuh App), challenging the platform's stability due to its non-standard distribution.
- **Monitored Endpoint: Windows 10 x64 VM:** Represents a typical high-value endpoint in a production environment, generating essential security logs (Event IDs) for monitoring.
- **Virtualization: VMware Workstation Pro:** Utilized to host and manage the isolated virtual lab environment, ensuring network segregation and easy snapshot management.

Network Topology

The virtual lab environment consists of two virtual machines connected via a NAT or Host-only virtual network, ensuring stable communication between the agent and manager. This setup isolates the security testing environment from the host machine and external networks.

- **Diagram:**



- **Communication Channels:** The diagram illustrates the critical data flows:
 - **Agent Data:** Logs and events are transmitted to the Manager over TCP Port 1514.
 - **Enrollment:** Key exchange for agent registration occurs over TCP Port 1515.
 - **Management:** Web access to the Dashboard is secured over TCP Port 443.

- **IP Addressing:**
 - Wazuh Manager (Kali): 192.168.153.128
 - Wazuh Agent (Windows 10): 192.168.153.130

3. Implementation Steps: Deployment and Configuration

Wazuh Manager Deployment (Kali Linux VM)

The Manager, Indexer, and Dashboard were deployed on the Kali Linux VM as an all-in-one installation using the Wazuh installation assistant script.

1. **Add Wazuh Repository Key:** The GPG key for the Wazuh package repository was added to the Kali VM.
2. **Download Installation Script:** The official wazuh-install.sh script for the matching v4.x series (specifically v4.14.0) was downloaded using wget.
3. **Execute All-in-One Installation:** The installation was executed with administrative privileges (sudo) to deploy all core components simultaneously:

```
Bash
sudo bash wazuh-install.sh --all-in-one --overwrite
```

4. **Firewall Configuration:** After installation, the necessary ports on the Kali VM were opened using ufw to allow communication from the Windows 10 Agent and dashboard access:
 - Agent Communication (Data): Port 1514/tcp
 - Agent Enrollment (Key Exchange): Port 1515/tcp
 - Dashboard Access: Port 443/tcp
5. **Access Dashboard:** The Dashboard was accessed via the Manager's IP address (<https://192.168.153.128>) using the generated admin credentials.

Wazuh Agent Deployment (Windows 10 VM)

The Wazuh Agent was installed on the Windows 10 VM, configured to point to the Kali Manager.

1. **Download Agent Installer:** The Windows Agent installer (wazuh-agent-4.14.0-1.msi) was downloaded onto the Windows 10 VM.

2. **Execute Silent Installation:** The installer was run using PowerShell with Administrator privileges, specifying the Manager's IP address and a clear agent name for easy identification:

PowerShell

```
.\wazuh-agent-4.14.0-1.msi /q WAZUH_MANAGER="192.168.153.128"  
WAZUH_AGENT_NAME="Windows10VM"
```

3. **Start Agent Service:** The Wazuh Agent service was immediately started to initiate the connection and enrollment process:

PowerShell

```
Start-Service wazuhsvc
```

4. **Configuration Verification:** The Agent's configuration file (C:\Program Files (x86)\ossec-agent\ossec.conf) was confirmed to correctly list the Manager's IP address (192.168.153.128) and the communication port (1514).

Successful Enrollment and Monitoring

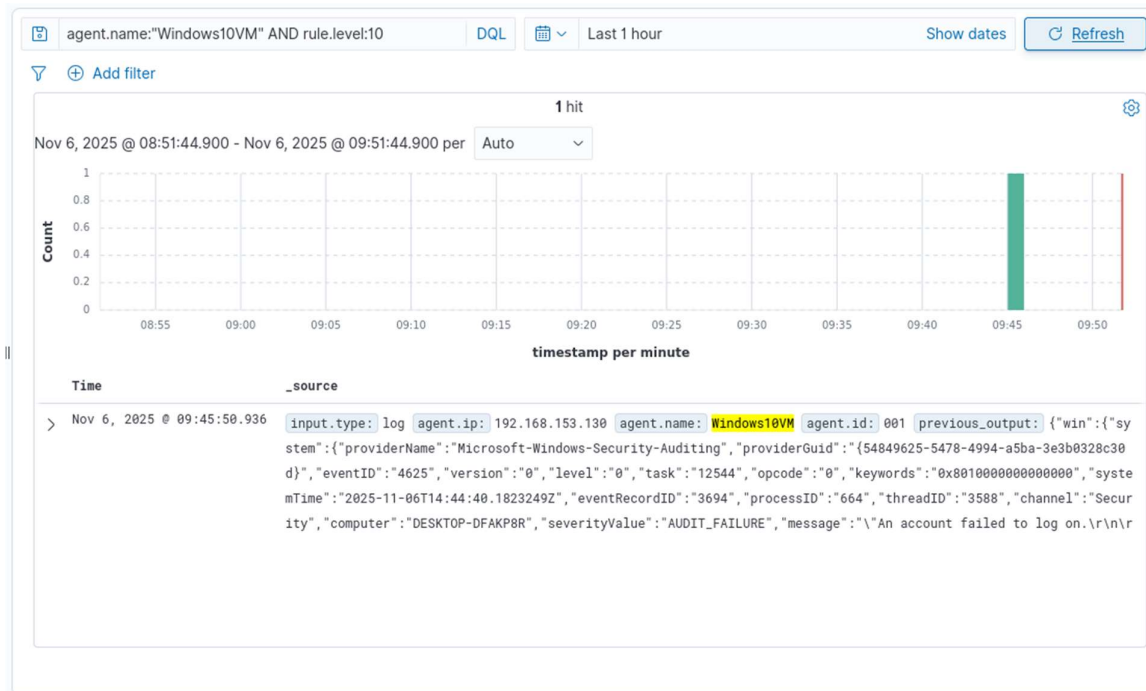
The Windows 10 Agent successfully connected and registered with the Kali Manager, transitioning to an **"Active"** status in the Wazuh Dashboard, confirming bidirectional communication and data flow for monitoring.

4. Security Detection Scenarios

Scenario A: Log Correlation (Brute Force Detection)

This scenario demonstrates the critical SIEM function of correlating multiple low-level events into a high-priority incident, proving that the system can distinguish between user error and malicious attack.

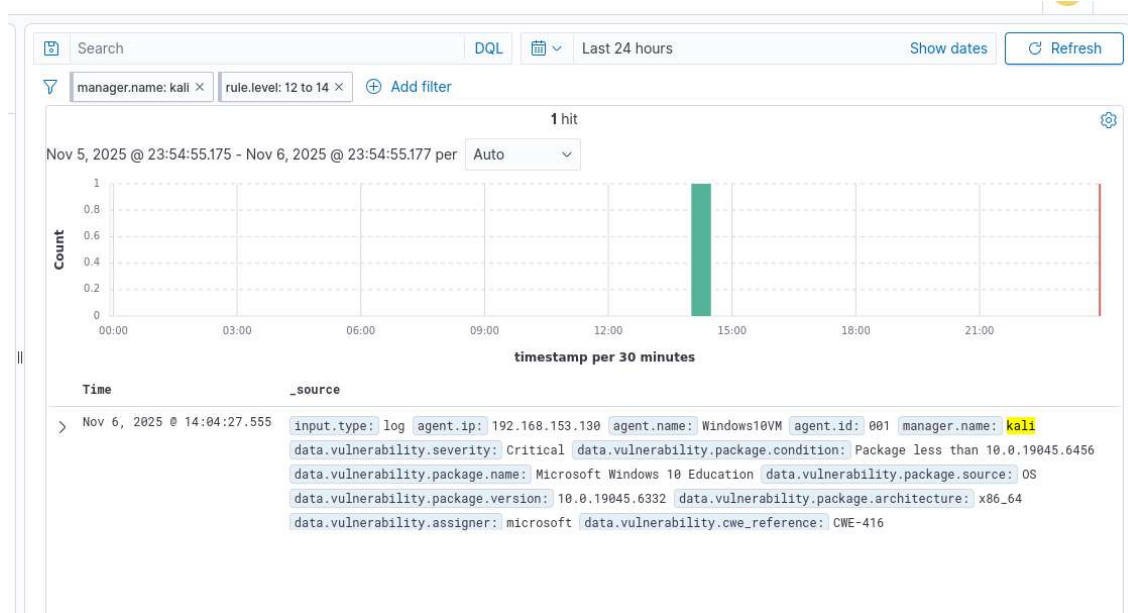
- Objective: Verify the Wazuh Manager's capability to correlate sequential Windows Event ID 4625 (Audit Failure) events into a single, high-severity incident.
- Test Action: Detail the steps taken on the Windows 10 VM (e.g., five failed login attempts in quick succession).
- Wazuh Analysis:
 - Detection: The Wazuh Manager's correlation engine detected the rapid frequency of failed logins.
 - Result: The system generated a High Severity (Rule Level 10+) alert, titled "Brute Force Attack Detected."
- Deliverable: Include the Screenshot of the correlated high-severity alert.



Scenario B: Vulnerability Detection (Critical Flaw)

This demonstrates proactive security monitoring by utilizing the SIEM's integrated vulnerability assessment tools.

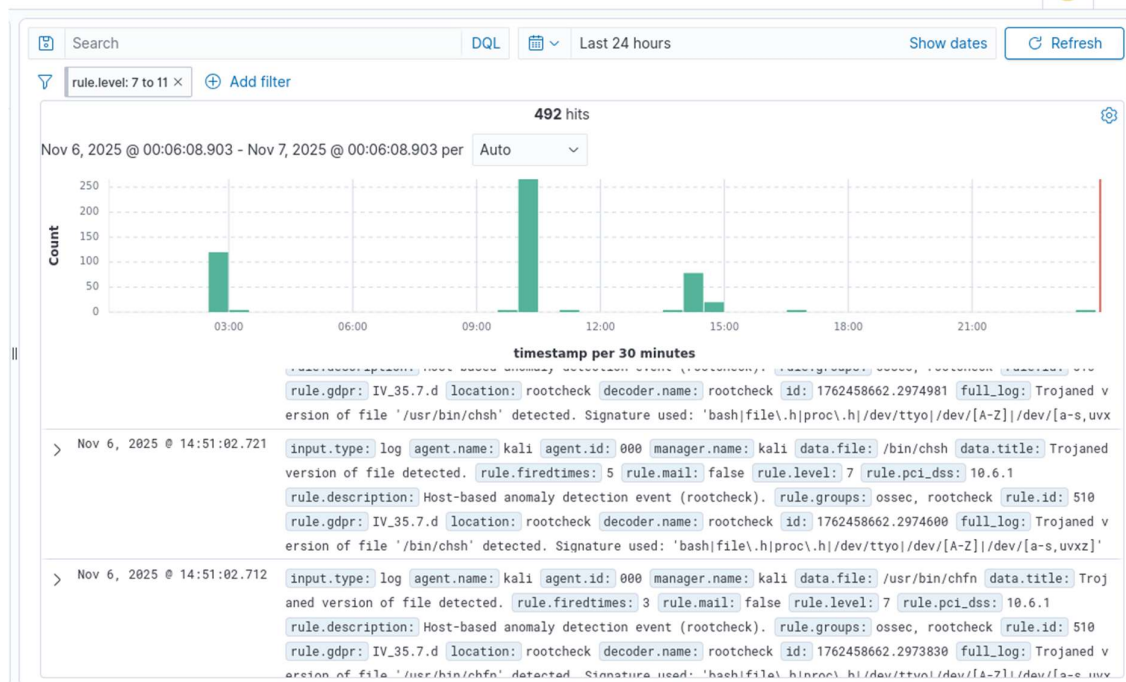
- Objective: Prove the system can automatically identify and alert on known software vulnerabilities on the monitored endpoint.
- Wazuh Analysis:
 - The Agent provided system inventory data (OS version) to the Manager. The Manager compared this data against the NVD (National Vulnerability Database).
 - Result: A Critical Severity (Level 12-14) alert was generated.
 - Finding: The Windows 10 operating system was detected with a critical flaw, specifically CWE-416 (Use After Free), a vulnerability often exploited for privilege escalation or remote code execution. This was detected because the installed package version was below the required security threshold.
- Remediation: State the necessary security action (immediately patching the Windows OS to the latest version)
- Deliverable: Include the Screenshot of the Critical Vulnerability Alert.



Scenario C: Host Integrity Monitoring (Internal HIDS)

This section demonstrates the comprehensive protection afforded by the SIEM, including the self-monitoring of the Manager host itself.

- Objective: Verify the integrity monitoring capability of the Manager's internal Rootcheck module.
- Test Action: Document that the Manager's scheduled HIDS scan ran automatically.
- Wazuh Analysis:
 - Detection: The Rootcheck module flagged a deviation in the digital signature of critical system binaries (/bin/chsh, /usr/bin/chfn).
- Result: A Medium Severity (Level 7) alert was generated, titled "Trojaned version of file detected" .
- Deliverable: Include the Screenshot of the Host-based anomaly detection alert.



5. Conclusion

- **Project Summary:** Reiterate that the Major Project successfully deployed and validated a functional SIEM platform (Wazuh v4.14.0).
- **Key Achievements:**
 - Confirmed the system's ability to perform log correlation (Brute Force)
 - Vulnerability assessment (CWE-416 detection)
 - Host Integrity Monitoring (Rootcheck alerts).
- **Technical Proficiency:** Emphasize that the successful operation was only achieved after extensive troubleshooting, including resolving critical version mismatches, complex firewall blocks, and inconsistent package states on the Manager OS, demonstrating high-level debugging and system administration skills.
- **Future Scope:** Suggest practical next steps, such as deploying active response rules (e.g., automatically blocking the source IP after a brute force attack) or integrating threat intelligence feeds.